XAI: Open Problems in Mechanistic Interpretability

FAMAF - UNC

First Semester 2026

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

🧉 Open Problems in Mechanistic Interpretability: A Whirlwind Tour ---

🧉 Motivation ---

Key Question

What should interpretability look like in a post GPT-4 world?

- Large, generative language models are a **big deal** - Models will keep scaling — what work done now will matter in the future? - Emergent capabilities keep arising - Many mundane problems go away - A single massive foundation model

🧉 Inputs and Outputs Are Not Enough ---

Goal: Understand Model Cognition

Is it aligned, or telling us what we want to hear?

What is a Transformer? ---

- **Input:** Sequences of words - **Output:** Probability distribution over the next word - **Residual stream:** A sequence of representations - One per input word, per layer - Each layer is an incremental update - Represents the word plus context - **Attention:** Moves information *between* words - Made of heads, each acts independently - **MLP:** Processes information *once* it's been moved to a word

🧉 What is Mechanistic Interpretability? ---

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- **Goal:** Reverse engineer neural networks - Like reverse-engineering a compiled binary to source code - **Hypothesis:** Models learn human-comprehensible algorithms and can be understood, if we learn how to make it legible - Understanding **features** — the variables inside the model - Understanding **circuits** — the algorithms learned to compute features - **Key property:** Distinguishes between cognition with identical output - A deep knowledge of circuits is crucial to understand, predict and **align** model behaviour

A Growing Area of Research ---

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🧉 Features = Variables: What Does the Model Know? ---

🧉 Features: Linear Representation of Quantities ---

🧉 Circuits = Functions: How Does the Model Think? ---

🧉 Induction Heads: A Key Circuit ---

🧉 Mechanistic Understanding of Induction Heads ---

Case Study: Emergence of In-Context Learning ---

🧉 The Mindset of Mechanistic Interpretability ---

- **Alien neuroscience:** Models *are* interpretable, but not in our language — if we learn to think like them, mysteries dissolve
- **Skepticism:** It's extremely easy to trick yourself in interpretability
  - **Zoom In:** Rigour and depth over breadth and scalability
- **Ambition:** It *is* possible to achieve deep and rigorous understanding
- A bet that models have underlying principles and structures that **generalise**

Personal Motivation (Neel Nanda)

Easy to get started, fast feedback loops, cross between maths, CS, natural sciences and truth-seeking. Code early and often — get contact with reality.

🧉 Case Study: Grokking ---

Mystery

🧉 The Modular Addition Circuit ---

🧉 Frontier: Polysemanticity & Superposition ---

Polysemanticity: One Neuron, Many Concepts ---

Hypothesis: Polysemanticity is due to Superposition ---

Geometry of Superposition ---

🧉 Case Study: Interpretability in the Wild ---

🧉 Refining Ablations: Mechanistic Interpretability as Validation ---

🧉 Technique: Activation Patching ---

🧉 Linear Representation Hypothesis ---

🧉 Case Study: Emergent World Representations in Othello-GPT ---

Othello-GPT: A Linear World Model ---

🧉 Learning More ---

- **200 Concrete Open Problems in Mechanistic Interpretability** neelnanda.io/concrete-open-problems - **Getting Started in Mechanistic Interpretability** neelnanda.io/getting-started - **A Comprehensive Mechanistic Interpretability Explainer** neelnanda.io/glossary - **TransformerLens** (Python library) github.com/neelnanda-io/TransformerLens

Open Problems covered in this talk

Studying Learned Features · Analysing Toy Language Models · Analysing Training Dynamics · Interpreting Algorithmic Models · Exploring Polysemanticity & Superposition · Finding Circuits in the Wild · Techniques, Tooling and Automation · Future Work on Othello-GPT

Thank You!

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